Homework 3: Research project proposal

1. Problem and Research Gap

Literature overview

Understanding how human capital specificity and skill structures govern mobility is a central challenge in economic and educational research. Neffke, Otto, and Weyh (2017) highlighted that although labor mobility plays an important role in economics, worker mobility patterns across industry boundaries and the inter-industry structure of labor flows have been largely neglected. Earlier relatedness measures – such as co-exports, co-occurrence, and input-output tables – often failed to capture the distance between industries in terms of human capital requirements and neglected inter-industry constraints to labor mobility.

Seeking to determine how industry-specific human capital is and which industries share similar skill requirements, Neffke et al. (2017) developed a novel labor-flow-based skill-relatedness index that significantly outperforms co-location and input output measures in predicting the entry and regional growth of local industries. Furthermore, they demonstrated that inter-industry labor flows are highly concentrated among a few industry pairs and form a network that is independent of a worker's occupation and geographic mobility (local or long-distance).

Subsequent literature extended this capability and network framework into new contexts, methods, and outcomes. Mealy and Teytelboym (2022) applied capability-based relatedness models to a dataset of traded green products, demonstrating how countries leverage existing productive capabilities for green diversification and carbon emissions reduction. In parallel, del Rio-Chanona et al. (2021) extended Neffke et al. (2017) by applying occupational mobility networks to analyse worker reallocation under automation shocks, developing a data-driven network model from occupational transition data to simulate macro-level labor market dynamics and micro-level unemployment outcomes across wage groups. Broader research on knowledge and collaboration networks (Abbasiharofteh et al., 2023; Fleming et al., 2007) likewise confirms that network clustering, path length, and brokerage govern the success of knowledge recombination.

Gap Statement

Previous work has shown that skill relatedness drives worker mobility across industries (Neffke et al., 2017) and that mobility networks shape reallocation outcomes (Del Rio-Chanona et al., 2021), but we do not know how skill-relatedness between undergraduate and postgraduate disciplines influences students' choices of master's programs and their subsequent academic performance. Addressing this gap matters because higher education institutions need data-driven insights to design flexible interdisciplinary curricula, optimize admission requirements, and reduce drop-out rates.

2. Research Questions, Hypotheses, and Network Mechanism

Research Questions and Hypotheses

How does the degree of skill-relatedness between undergraduate and postgraduate fields affect students' choices when selecting a master's program? To what extent does a higher skill relatedness predict academic success and program completion in master's studies? Building on these questions, we test two hypotheses:

- (H1) students are significantly more likely to select master's (MA/MSc) programs connected to their bachelor's (BA/BSc) field by strong skill-relatedness weights, short network path lengths, and shared disciplinary clusters;
- (H2) higher skill-relatedness and closer network proximity between undergraduate and postgraduate study fields positively predict academic performance (grade point average, GPA) and program completion rates while reducing drop-out probabilities.

Mechanism (Why Does the Network Matter?)

Dyadic comparisons between two isolated degree programs fail to capture indirect skill transferability across the wider academic ecosystem. The network structure matters because academic competencies form an interconnected system of complementary skills (Fleming et al., 2007; Abbasiharofteh et al., 2023). Even when a bachelor's and a master's program exhibit weak direct historical transition flows, a short weighted path length or high local clustering indicates shared foundational analytical and methodological competencies across neighboring disciplines, which lowers cognitive adaptation costs for transitioning students. Conversely, transitions across highly segregated network clusters with long path lengths lack intermediate bridging skills, creating structural bottlenecks and academic friction that increase drop-out risk (del Rio-Chanona et al., 2021).

3. Data and Method

Nodes, Links, and Data Sources

In our proposed network, nodes represent specific undergraduate (BA/BSc) and postgraduate (MA/MSc) degree programs. Directed, weighted links measure skill-relatedness in two complementary ways:

1) A revealed student transition network comparing observed BA-to-MA enrollment flows against baseline random expectations (Neffke et al., 2017).
2) A curriculum skill-overlap network measuring text similarity between skills provided in undergraduate syllabi and skills required in master's syllabi.

We utilize three data sources:

1) Administrative student transition records tracking admissions and enrollment flows from specific bachelor's programs to master's programs.
2) Course syllabi and curriculum descriptions mapping provided undergraduate skills and required postgraduate skills.
3) Academic performance metrics including course grades, cumulative GPA, and program completion rates.

Network Measures, Data Accessibility, and Plan B

To test our hypotheses, we compute four network measures:

1) Edge weight (skill-relatedness index) to capture direct competence proximity between degree programs.
2) Shortest weighted path length to measure indirect skill distance across intermediate study fields.
3) Clustering coefficient and modularity (segregation) to evaluate disciplinary cohesion versus cross-faculty silos.
4) Network robustness to simulate how introducing or removing interdisciplinary bridging courses alters connectivity between segregated study clusters.

Regarding data accessibility, course syllabi and program requirements are publicly accessible on university websites, while anonymized student transition and GPA records can be requested from university registries under academic data-sharing protocols. As a Plan B, if individual-level GPA microdata is inaccessible due to privacy restrictions, we will construct the skill-relatedness network entirely from publicly available course syllabi and evaluate transitions using publicly accessible aggregate admission matrices, program level completion statistics, and public educational trajectory profiles.

4. Expected Results and Impact

Expected Results

If our hypotheses hold, we expect to find that bachelor's-to-master's enrollment flows are heavily concentrated within tightly clustered, high-relatedness network neighbourhoods. Furthermore, shorter network path lengths and higher skill-relatedness between prior and current studies will be associated with significantly higher master's GPAs and completion rates, whereas students jumping across segregated clusters without bridging coursework will face elevated drop-out risk.

Expected Impact

This research delivers actionable value for three key groups:_

1) University leaders and academic decisionmakers gain empirical tools to detect structurally risky transition pathways, enabling early interventions that reduce institutional drop-out rates and improve graduation efficiency.
2) Program directors can use skill-relatedness maps to calibrate admission prerequisites, design flexible interdisciplinary curricula, and introduce targeted preparatory bridging modules for students entering from distant undergraduate fields.
3) Educational policy makers and students obtain system-level transparency into disciplinary segregation, supporting evidence-based curriculum reform and better-informed master's study choices.

References:

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